

A block-projection counterexample for complex interpolation of weighted Sobolev spaces

Autonomous proof attempt for run fa_banach_001

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Status. Candidate full affirmative answer to the first question in Section 8.3 of Cwikel–Einav, together with a stronger counterexample to the unrestricted Stein–Weiss-type identity. The proof is analytic and is submitted for human review.

Abstract

We construct positive compact-bounded weights w_0, w_1 on $\mathbb{R}$ for which $r = w_0/w_1$ is locally Lipschitz and, at $p = 2$ and $\theta = 1/2$,

$$\mathcal{W}^2(\mathbb{R}, 1/2, r) \subsetneq [W^{1,2}(\mathbb{R}, w_0), W^{1,2}(\mathbb{R}, w_1)]_{1/2} \subsetneq W^{1,2}(\mathbb{R}, \sqrt{w_0 w_1}).$$

The first strict inclusion answers the exact open question. The second gives a counterexample to the unrestricted weighted-Sobolev analogue of the Stein–Weiss formula. The mechanism is a common complemented block subspace: its endpoint norm is equivalent to unweighted ℓ^2 , while its geometric-mean norm is exponentially smaller.

1 Source question

The source is M. Cwikel and A. Einav, *Interpolation of weighted Sobolev spaces*, arXiv:1807.10790, later published in the *Journal of Functional Analysis* 277 (2019), 2381–2441 [1]. On PDF page 47, at the start of Section 8.3, the authors ask:

Can one find an open set U and weights w_0, w_1 on it for which

$$\mathcal{W}^p(U, \theta, r_w) \subsetneq [W^{1,p}(U, w_0), W^{1,p}(U, w_1)]_\theta?$$

For positive weights satisfying the source paper’s compact boundedness condition, write

$$X_j = W^{1,2}(\mathbb{R}, w_j), \quad r = \frac{w_0}{w_1}, \quad w_{1/2} = \sqrt{w_0 w_1}.$$

The auxiliary space in the question is

$$\mathcal{W}^2(\mathbb{R}, 1/2, r) = \{u \in W^{1,2}(\mathbb{R}, w_{1/2}) : u |(\log r)'| \in L^2(\mathbb{R}, w_{1/2})\}.$$

Theorem 1.21 of [1] says that on $U = \mathbb{R}$, if w_0, w_1 are compact-bounded and r is locally Lipschitz, then

$$\mathcal{W}^2(\mathbb{R}, 1/2, r) \subset [X_0, X_1]_{1/2} \subset W^{1,2}(\mathbb{R}, w_{1/2}). \tag{1}$$

Using (7.9) yields the inequality required to complete the proof of this theorem. $\square$

8.3. Future research and open problems. We feel excited about future possibilities in continuing to try and understand whether or not one can obtain a Stein-Weiss like theorem for weighted Sobolev spaces under more general hypotheses on the underlying set and the weight functions. In particular, some of the problems which we think are worth investigating are the following:

Question. Can one find an open set U and weights ω_0 and ω_1 on it for which

$$\mathcal{W}^p(U, \theta, r_\omega) \subsetneq [W^{1,p}(U, \omega_0), W^{1,p}(U, \omega_1)]_\theta?$$

Question. Can one find necessary and sufficient conditions under which

$$W_0^{1,p}(U, \omega) = W^{1,p}(U, \omega) \quad \text{and} \quad \mathcal{W}_0^p(U, \theta, r_\omega) = \mathcal{W}^p(U, \theta, r_\omega)?$$

Let us mention that pages 550-554 of [12] contain a discussion of the closure of $C_c^\infty(U)$ (there denoted by $C_0^\infty(U)$) in various weighted Sobolev spaces. A different but somewhat related topic is considered on pages 43-49 of [11].

Figure 1: The question as rendered from PDF page 47 of arXiv:1807.10790.

2 Idea of the proof

In the n th unit cell we place a very short core of length M_n^{-1} . Across that core the two weights exchange the values M_n and 1 in a reflection-symmetric way. A block function is constant on the core and tapers to zero through two fixed-width collars. Both weights are multiplied by the tiny common factor M_n^{-4} on the collars. This common factor makes the taper cheap but cancels from $r = w_0/w_1$.

The short core has two consequences. First, the endpoint norm of one block stays comparable to one. Second, averaging over the core is uniformly bounded on both endpoint Sobolev spaces: a function cannot change its core average cheaply because the core is short. These averages define a single bounded projection on both endpoints. Reflection makes the projected ranges identical Hilbert sequence spaces.

For the first strict inclusion, coefficients $1/n$ put a block sum in both endpoints, hence in the interpolation space. The rapid exchange of the weights forces $|(\log r)'| \asymp M_n \log M_n$ on the core, so the auxiliary norm diverges. For the stronger strict inclusion, coefficients $n^{-1/2}$ are square summable for the exponentially small geometric-mean block norm, but not for the projected endpoint norm. The common projection then rules out membership even in $X_0 + X_1$.

3 Construction

For $n \geq 1$ set

$$C_n = (n, n+1), \quad z_n = n + \frac{1}{2}, \quad M_n = 16^n, \quad L_n = M_n^{-1}, \quad h = \frac{1}{8},$$

and let

$$a_n = z_n - \frac{L_n}{2}, \quad b_n = z_n + \frac{L_n}{2}, \quad I_n = (a_n, b_n).$$

Define a continuous piecewise affine profile $s_n : C_n \rightarrow [0, 1]$ by

$$s_n(x) = \begin{cases} \frac{1}{2}, & x \notin (a_n - h, b_n + h), \\ \frac{1}{2} + \frac{x - (a_n - h)}{2h}, & a_n - h < x < a_n, \\ 1, & a_n \leq x \leq a_n + L_n/3, \\ 2 - 3\frac{x - a_n}{L_n}, & a_n + L_n/3 < x < a_n + 2L_n/3, \\ 0, & a_n + 2L_n/3 \leq x \leq b_n, \\ \frac{x - b_n}{2h}, & b_n < x < b_n + h. \end{cases} \quad (2)$$

Let the common factor be

$$q_n(x) = \begin{cases} 1, & x \in I_n, \\ M_n^{-4}, & x \in C_n \setminus I_n, \end{cases}$$

and, for $x \in C_n$, put

$$w_0(x) = q_n(x)M_n^{s_n(x)}, \quad w_1(x) = q_n(x)M_n^{1-s_n(x)}. \quad (3)$$

At points outside the union of the open cells, set $w_0 = w_1 = 1$.

The n th block is the symmetric tent

$$\beta_n(x) = \begin{cases} 0, & x \notin (a_n - h, b_n + h), \\ \frac{x - (a_n - h)}{h}, & a_n - h < x < a_n, \\ 1, & a_n \leq x \leq b_n, \\ \frac{b_n + h - x}{h}, & b_n < x < b_n + h. \end{cases} \quad (4)$$

The supports of the β_n are pairwise disjoint.

Lemma 1 (admissibility). *The functions w_0, w_1 are strictly positive measurable weights satisfying the compact boundedness condition. Their quotient $r = w_0/w_1$ is locally Lipschitz, and on C_n*

$$r = M_n^{2s_n-1}, \quad w_{1/2} = \sqrt{w_0 w_1} = q_n M_n^{1/2}.$$

Proof. Every compact subset of $\mathbb{R}$ meets only finitely many cells. On each of those cells each weight assumes only finite positive values bounded above and away from zero, so compact boundedness follows. The common factor cancels from the quotient. Since $s_n = 1/2$ near both endpoints of C_n , the cellwise quotient joins the constant value 1 across cell boundaries. It is piecewise Lipschitz in each cell; on a compact set only finitely many pieces occur. This proves local Lipschitz continuity. The formula for $w_{1/2}$ is immediate from (3). $\square$

Let $R_n x = 2z_n - x$. Directly from the construction,

$$s_n(R_n x) = 1 - s_n(x), \quad q_n(R_n x) = q_n(x), \quad \beta_n(R_n x) = \beta_n(x). \quad (5)$$

4 The common complemented block subspace

Define

$$B_{j,n} = \|\beta_n\|_{X_j}^2 = \int_{C_n} w_j(|\beta_n|^2 + |\beta'_n|^2), \quad j = 0, 1.$$

Lemma 2 (block estimates). *For every n ,*

$$B_{0,n} = B_{1,n} =: B_n, \text{quad} \frac{1}{3} \leq B_n \leq 1 + \frac{65}{4} M_n^{-3} < 1.005. \quad (6)$$

Moreover, if

$$D_n = \|\beta_n\|_{W^{1,2}(\mathbb{R}, w_{1/2})}^2,$$

then

$$D_n \leq M_n^{-1/2} + \frac{65}{4} M_n^{-7/2}. \quad (7)$$

Proof. Equation (5) exchanges w_0 and w_1 and fixes $|\beta_n|^2 + |\beta'_n|^2$, so $B_{0,n} = B_{1,n}$. On the core, $\beta_n = 1$, $\beta'_n = 0$, and an exact calculation gives

$$\begin{aligned} \int_{I_n} w_0 &= L_n \left(\frac{M_n}{3} + \frac{M_n - 1}{3 \log M_n} + \frac{1}{3} \right) \\ &= \frac{1}{3} + \frac{M_n - 1}{3 M_n \log M_n} + \frac{1}{3 M_n}. \end{aligned}$$

This lies between $1/3$ and 1 . The same holds for w_1 by reflection. On the two collars, $w_j \leq M_n^{-3}$, $|\beta_n| \leq 1$, and $|\beta'_n| = h^{-1} = 8$. Their total contribution is at most

$$2h(1 + h^{-2})M_n^{-3} = \frac{65}{4} M_n^{-3},$$

which proves (6). For $w_{1/2}$, the core contribution is $L_n M_n^{1/2} = M_n^{-1/2}$ and the collar weight is $M_n^{-7/2}$; the same calculation gives (7). $\square$

For a locally Sobolev function f , let

$$m_n(f) = \frac{1}{L_n} \int_{I_n} f(x) dx, \quad Pf = \sum_{n \geq 1} m_n(f) \beta_n.$$

Lemma 3 (common projection). *The same formula P defines a bounded projection on both X_0 and X_1 . Its range in either endpoint is*

$$\mathcal{R} = \left\{ \sum_{n \geq 1} c_n \beta_n : \sum_{n \geq 1} B_n |c_n|^2 < \infty \right\}. \quad (8)$$

In particular, the two endpoint range spaces coincide isometrically.

Proof. Fix $j \in \{0, 1\}$. Let $H_{j,n}$ be the third of I_n on which $w_j = M_n$: the left third for $j = 0$ and the right third for $j = 1$. Weighted Sobolev functions are absolutely continuous on the compact core. Comparing the two averages and using the fundamental theorem of calculus gives

$$|m_n(f)| \leq \left| \frac{3}{L_n} \int_{H_{j,n}} f \right| + \int_{I_n} |f'|.$$

Cauchy–Schwarz, $M_n L_n = 1$, and $w_j \geq 1$ on I_n yield

$$\begin{aligned} |m_n(f)|^2 &\leq \frac{6}{M_n L_n} \int_{H_{j,n}} w_j |f|^2 + 2L_n \int_{I_n} w_j |f'|^2 \\ &\leq 6 \int_{I_n} w_j (|f|^2 + |f'|^2). \end{aligned}$$

The supports of the blocks are disjoint, so Lemma 2 implies

$$\|Pf\|_{X_j}^2 = \sum_n B_n |m_n(f)|^2 \leq 6 \sup_n B_n \|f\|_{X_j}^2.$$

Since $m_n(\beta_k) = \delta_{nk}$, $P^2 = P$. Disjointness and $B_{0,n} = B_{1,n} = B_n$ now give exactly (8) at both endpoints. $\square$

5 Both strict inclusions

Theorem 4. *For the weights (3),*

$$\mathcal{W}^2(\mathbb{R}, 1/2, r) \subsetneq [X_0, X_1]_{1/2} \subsetneq W^{1,2}(\mathbb{R}, w_{1/2}).$$

Consequently, the answer to the first question in Section 8.3 of [1] is affirmative, and the unrestricted identity

$$[W^{1,2}(\mathbb{R}, w_0), W^{1,2}(\mathbb{R}, w_1)]_{1/2} = W^{1,2}(\mathbb{R}, \sqrt{w_0 w_1})$$

is false even when both weights are compact-bounded and their quotient is locally Lipschitz.

Proof. For the first strictness, set

$$g = \sum_{n \geq 1} \frac{1}{n} \beta_n.$$

By (6), $\|g\|_{X_j}^2 = \sum_n B_n/n^2 < \infty$ for $j = 0, 1$. Hence $g \in X_0 \cap X_1 \subset [X_0, X_1]_{1/2}$.

On the middle third of I_n , $\beta_n = 1$ and

$$|(\log r)'| = \frac{6 \log M_n}{L_n}, \quad w_{1/2} = M_n^{1/2}.$$

Therefore that third alone contributes

$$\frac{1}{n^2} \frac{36(\log M_n)^2}{L_n^2}, M_n^{1/2} \frac{L_n}{3} = \frac{12}{n^2} (\log M_n)^2 M_n^{3/2} \quad (9)$$

to $\int |g|^2 |(\log r)'|^2 w_{1/2}$. These terms do not even tend to zero, so $g \notin \mathcal{W}^2(\mathbb{R}, 1/2, r)$. Lemma 1 permits us to apply the source inclusion (1); hence the first inclusion is strict.

For the second strictness, define

$$f = \sum_{n \geq 1} \frac{1}{\sqrt{n}} \beta_n.$$

By (7),

$$\|f\|_{W^{1,2}(w_{1/2})}^2 \leq \sum_{n \geq 1} \frac{1}{n} \left(M_n^{-1/2} + \frac{65}{4} M_n^{-7/2} \right) < \infty.$$

Thus $f \in W^{1,2}(\mathbb{R}, w_{1/2})$.

Suppose, toward a contradiction, that $f \in X_0 + X_1$. The common formula for P , bounded on each endpoint, defines a bounded map from the algebraic sum into $\mathcal{R} + \mathcal{R} = \mathcal{R}$. But $Pf = f$, because every block is identically one on its own averaging core. Hence (8) would give

$$\sum_{n \geq 1} \frac{B_n}{n} < \infty,$$

contradicting $B_n \geq 1/3$. Therefore $f \notin X_0 + X_1$. Every complex interpolation space is an intermediate space contained in the endpoint sum, so $f \notin [X_0, X_1]_{1/2}$. The second inclusion in (1) is strict. $\square$

6 Verification, limitations, and review focus

The companion script `code/verify_blocks.py` checks the reflection identities on 12,006 sample points, the exact core-mass formula, the uniform endpoint and projection bounds, the geometric-mean decay, and finite partial sums for both witnesses. It is a sanity check; the proof above uses the displayed analytic inequalities.

A bounded novelty search through 13 August 2026 used the run's four cheap indexes, local arXiv full-text sources and citation traces, and exact web/arXiv queries for the source question and close variants. Later related work included arXiv:2204.11583 on density and power weights on the half-line and arXiv:2503.14636 on complex interpolation of power-weighted Sobolev spaces with boundary conditions. Neither contained this block construction, an answer to the strict-inclusion question, or a general counterexample of this form. This is evidence only, not a claim of exhaustive novelty.

The construction deliberately uses measurable weights with jumps in the common factor q_n ; this is allowed by the source's compact boundedness hypothesis. The quotient w_0/w_1 , which is the quantity required to be locally Lipschitz, is continuous and locally Lipschitz. Human review should focus on the boundedness and common-range assertions for P , and on the application of Theorem 1.21 of [1] to pass from the full auxiliary space to the interpolation space on $\mathbb{R}$.

References

- [1] M. Cwikel and A. Einav, *Interpolation of weighted Sobolev spaces*, J. Funct. Anal. 277 (2019), no. 7, 2381–2441; arXiv:1807.10790.